

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Calculus II

Semester: Fall

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Evaluate integral: $\int_1^2 \int_0^4 2xy \, dy \, dx$. 5
- b) State Dirichlet's Theorem. Using it computes 5
$$\iiint x^2 yz \, dx \, dy \, dz$$
over the volume of the tetrahedron bounded by $x = 0, y = 0, z = 0$ and $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.
- c) Find the volume bounded by xy plane, the cylinder $x^2 + y^2 = 1$, 5
 $x + y + z = 3$ and the plane.
2. a) Solve by using power series method: 7
 $y'' - 4xy' + (4x^2 - 2)y = 0$ by power series method solution.
- b) Define the Legendre's differential equation and find its *general* 8
solution.

OR

Write the Bessel's functions of order n and $-n$ and hence show that:

i. $J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$

ii. $J_{-\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \cos x$

3. a) Solve the initial value problem by using Laplace Transform. 7
 $y'' + 2y' + 5y = 0, y(0) = 2, y'(0) = -4$.
- b) Define unit step function of Laplace transforms. 8
 - i. Find the Laplace transform of $te^{-t} \cos ht$
 - ii. Find the inverse Laplace transform of $\log \frac{s(s+1)}{s^2+4}$

4. a) The acceleration of a particle at any time t is 5
 $\vec{a} = 9 \cos 3t \vec{i} - 4 \sin 2t \vec{j} + 6t \vec{k}$. If the particle starts to move from rest, find the velocity and displacement at any time t .
- b) Find the directional derivative of $f = x^2yz + 4xz^2$ at $(1, -2, 1)$ in the direction $2\vec{i} - \vec{j} - 2\vec{k}$. 5
- c) Using Green's theorem, evaluate $\oint_C [(2x^2 - y^2) dx + (x^2 + y^2) dy]$ where C is the boundary of the semi-circular disk $x^2 + y^2 \leq 1$, $y \geq 0$. 5
5. a) Evaluate: $\iint_S \vec{F} \cdot \vec{n} dA$ where; $\vec{F} = (x^2, y^2, z^2)$, $S: x + y + z = 1$, $x \geq 0, y \geq 0, z \geq 0$. 8
- b) Using Stoke's theorem, Evaluate $\oint_C [4y dx + 2z dy + 6y dz]$ where C is the curve of intersection of the sphere $x^2 + y^2 + z^2 = 6z$ and the plane $z = x + 3$. 7

OR

Use Gauss divergence theorem to evaluate $\iiint_S \vec{F} \cdot \hat{n} ds$, where

$\vec{F} = 4x\vec{i} - 2y^2\vec{j} + z^2\vec{k}$ and S is the surfaces bounding the region $x^2 + y^2 \leq 4$, $z = 0$ and $z = 3$.

6. a) State whether the given function is even or odd. Find its Fourier series 8

$$f(x) = \begin{cases} k, & \text{if } -\frac{\pi}{2} < x < \frac{\pi}{2} \\ 0, & \text{if } \pi/2 < x < 3\pi/2 \end{cases}$$

And show that $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}$.

- b) Find the Fourier sine and cosine series of $f(x) = e^x$ for $0 \leq x \leq l$. 7

7. Write short notes on: (Any two) 2×5

- a) Derive the equation for the shock path
- b) Solve the advection equation: $u_t + 2uu_x = 0$, $u(x, 0) = e^{-x}$.
- c) If $f(x)$ is even function then prove that

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$$